Victor Li

li.victor@northeastern.edu | (302) 397-7170 | Boston, MA
linkedin.com/in/victor-li | vvictorrr.github.io | github.com/vvictorrr
Co-op Availability: July - December 2026

Summary

Data Science student and AI/ML engineer with production experience building LLM/RAG systems, machine learning models for scientific data, and cloud-native applications. Research background spans particle physics, materials science, and AI security, with a 4.00 GPA and a first-author peer-reviewed publication.

Education

Northeastern University, Boston, MA *Expected May 2028*
Candidate for B.S., Artificial Intelligence; Minors: Physics, Mathematics | GPA: 4.00/4.00
Related Coursework: Advanced Programming with Data, Foundations of Data Science, Algorithms and Data, Foundations of Cybersecurity, Databases, Programming in C++, Probability and Statistics, Linear Algebra
Honors: AJC Merit Research Scholar, CERN (selected from 450+ applicants); Dean's List; MCAM x FYW Oakland Art & Writing Contest

Publications

- **V. Li**, R. M. Slezak, L. T. Schmidt, D. Heiman, and H. Kumarakuru, "A Modular Electronic Locking System Utilizing Hexadecimal Input and Logic-Based Verification," *2026 ASEE Northeast Section Conference*, Southern New Hampshire University, Mar. 2026. doi: 10.18260/1-2–58569 *(first author)*
- L. R. Kempton, C. Smith, R. M. Slezak, **V. Li**, K. Zhang, D. Heiman, and H. Kumarakuru, "From Circuits to Curiosity: Building a Digital Reaction-time Electronic Game for Engaging Students in Experimental STEM Education," *2026 ASEE Northeast Section Conference*, Southern New Hampshire University, Mar. 2026. doi: 10.18260/1-2–58587
- **V. Li** et al., "Crystal structure of..." *Acta Crystallographica Section E*, 2024. Results presented at the Spring 2024 American Chemical Society National Meeting. *(first author)*

Technical Skills

- **Languages:** Python, SQL, R, C++, JavaScript, C
- **Machine Learning & AI:** PyTorch, TensorFlow, scikit-learn, XGBoost, NLP, LLM/RAG pipelines (retrieval-augmented generation, embeddings, hallucination evaluation, regression testing), Anthropic API / OpenAI API integration, Pandas, NumPy
- **Vector Search & Databases:** ChromaDB, MySQL, PostgreSQL, Apache Derby
- **Cloud, DevOps & Infrastructure:** AWS (Lambda, EventBridge, SNS, IAM, CloudWatch, DynamoDB), Docker, GitLab CI/CD, Serverless Architecture, Boto3, Conda/Mamba
- **Scientific Computing:** PyROOT, HTCondor (batch/grid processing), Jupyter Notebook, Web Scraping (Beautiful Soup)
- **Web & Tools:** Flask, RESTful APIs, React, HTML/CSS, Git/GitHub, VS Code, PyCharm, PROS

Research & Technical Experience

CMS Experiment @ CERN *July 2026 - Dec. 2026*
AJC Merit Research Scholar, AI/ML Engineer *Geneva, Switzerland*
- Built a generalizable RAG documentation pipeline (ingestion, retrieval, LLM Q&A, validation/regression testing) for CERN QC and research documentation, plus a reusable API-based tool for mining institutional archives at scale
- Built and completed a full CMS MET-filter data/MC analysis pipeline (PyROOT, HTCondor batch processing, jet-ID/veto-map and pileup corrections, cross-section-weighted MC stitching) supporting a dark matter search analysis
- Performed anomaly detection on data from the High Granularity Calorimeter (HGCAL) to support detector quality assurance, electronics monitoring, and searches for rare physics signatures

SpringLab at Northeastern *Mar. 2026 - Jun. 2026*
Undergraduate Research Volunteer *Boston, MA*
- Maintained and improved phototherapy data collection systems; enhanced an LED-photodiode array for uniform cell imaging; supported robot arm integration and contributed to electronics and PCB design
- Developed Python tools and a GUI to streamline data acquisition and support experimental automation

Oak Ridge National Laboratory *Sept. 2024 - Jan. 2025*
Undergraduate Research Assistant *Remote*
- Collaborated with Prof. Miguel Fuentes-Cabrera and a small team to develop machine learning models for interpreting neutron reflectivity spectra of thin-film polymers, materials critical to next-generation energy storage
- Built and tested diffusion-based models, applying PCA, multilayer perceptrons, and autoencoders to analyze experimental data generated at the Spallation Neutron Source

Harker ISE Lab, University of Delaware *Jun. 2023 - Aug. 2023*
Research Assistant *Newark, DE*
- First author on a peer-reviewed publication in *Acta Crystallographica Section E*; results presented at the Spring 2024 American Chemical Society National Meeting

- Conducted crystallography research under Prof. Chaoying Ni, designing experiments, analyzing data, and producing publication-ready documentation
- Analyzed and validated the structure of a new potential anticancer drug

Stability AI *Jul. 2023 - Sept. 2023*

LLM Security Researcher *Remote*

- Contributed to a White House-backed AI cybersecurity program at Stability AI, with results shared at DEF CON
- Conducted red-teaming of large language models, identified critical flaws, and designed mitigations adopted by the security team to strengthen model robustness against adversarial attacks

Projects

HEP-RAG - Physics Literature Q&A / *Python, SPECTER2, ChromaDB, DeBERTa-v3, Anthropic API* *Aug 2026*

- Built a RAG system over 80 arXiv HEP preprints (hep-ex/hep-ph) with hybrid retrieval (BM25 + SPECTER2 dense + reciprocal rank fusion + cross-encoder reranking); ran a 20-question hand-labeled ablation study across 4 retrieval configs, diagnosing a domain-mismatch failure mode where dense-only achieved 0% Recall@5 versus 93.3% for the full hybrid+reranked pipeline
- Implemented a dual-signal hallucination detection architecture (LLM-as-judge + DeBERTa-v3 NLI cross-encoder) achieving 0.59/0.97/0.74 precision/recall/F1; dual-signal detector caught a live citation-misattribution error the LLM judge alone missed; deployed as a Streamlit app on Streamlit Community Cloud

Automated DCA Investment Optimizer / *Python, AWS Lambda, DynamoDB, EventBridge, SNS, Boto3* *Jan. 2026*

- Designed a serverless investment timing system using multi-timeframe analysis and AWS event-driven architecture; Lambda functions perform hourly ETF price monitoring during investment windows, storing data in DynamoDB and triggering SNS alerts based on volatility-adjusted buy signals

Battery Prognostics Modeling / *Python, Pandas, NumPy, scikit-learn, XGBoost, Matplotlib* *Nov. 2025 - Dec. 2025*

- Developed predictive machine learning models for lithium-ion battery capacity fade and remaining useful life (RUL) using NASA battery aging datasets, addressing time-series correlation and data leakage issues
- Engineered physics-based features from voltage, current, temperature, and impedance data and evaluated multiple regression models, achieving $R^2 \approx 0.94$ for capacity prediction and $R^2 \approx 0.97$ for RUL prediction

Hexa-Quad-Tri Lock / *Digital Logic ICs, Flip-Flops, 555 Timer, FSM Design, Breadboards* *Oct. 2025 - Mar. 2026*

- Designed and implemented a hexadecimal combination lock supporting four-digit, four-bit inputs (~65,000 combinations) with a three-attempt lockout mechanism
- Implemented gate-level comparison logic, a Johnson counter for failed attempts, and debounce circuitry (Schmitt trigger, 555 timer); published and presented at the 2026 ASEE Northeast Section Conference. doi: 10.18260/1-2–58569

Debt Tracker / *Python, Flask, PostgreSQL, HTML/CSS/JS, Pandas, Matplotlib* *Jul. 2025 - Aug. 2025*

- Designed and built a full-stack web application to track shared expenses, simplify debt networks, and reduce transaction fees by minimizing unnecessary payments using a Flask backend and PostgreSQL database
- Used Pandas and Matplotlib to analyze real-world testing data from a sample group of four: reduced 30 payments to 3 and avoided $17.57 in PayPal fees, compounding to $913.64 per year

Work Experience

Wilmington Pharmatech *May 2026 - June 2026*

Data Analyst *Christiana, DE / Remote*

- Led a data quality initiative to clean and restructure a 32,000+ contact HubSpot CRM database
- Built automated Python tools to classify contacts by job title relevance, detect duplicate and impersonated records, normalize company naming inconsistencies, and cross-reference contacts against business development priorities
- Integrated the RocketReach API to enrich incomplete records and collaborated with BD leadership to define filtering criteria aligned with the company's ideal customer profile
- Reduced the database from 32,000+ contacts and 4,500+ companies to ~10,000 high-quality contacts and ~600 target companies, eliminating over 69% of outdated, irrelevant, or invalid records

Summer Learning Collaborative *Jun. 2024 - Aug. 2024*

Reading Coach *Wilmington, DE*

- Taught reading and literacy skills tailored to individual student needs and assessed progress throughout a 5-week program
- Delivered measurable improvements in student reading performance across participating students

Wilmington Pharmatech *Jun. 2022 - Sept. 2022*

IT Intern *Christiana, DE*

- Assisted the IT Manager in upgrading computers, networks, and security systems; provided employee technical support; improved secure communication and cross-platform data sharing; authored concise system manuals
- Increased system reliability, reduced support issues, and streamlined onboarding for future employees

Other Experience

NURobotics *Sept. 2025 - Present*

Programmer and Instructor *Boston, MA*

- Programmer for the World Championship qualifying VEXU Robotics team; developed a C++ pure-pursuit path-following algorithm with onboard odometry and contributed to a web-based autonomous pathing UI
- Teach the club's internal Intro to Robotics course, guiding new members through the full hardware-software design process using CAD, 3D printing, and Arduino programming

FIRST Robotics *Sept. 2022 - May 2024*

Programming Lead, Team 22730 *Wilmington, DE*

- Programming Lead and Robot Pilot for two years for a First Tech Challenge team; team placed 2nd in the State and won the Judge's Choice Award and State Finalist Award
- Used TensorFlow to train an onboard image recognition model to automate behavior based on randomized circumstances